

Lab 4: SQL Querying Solutions

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Database 1: Olympic Summer Games

1) RM

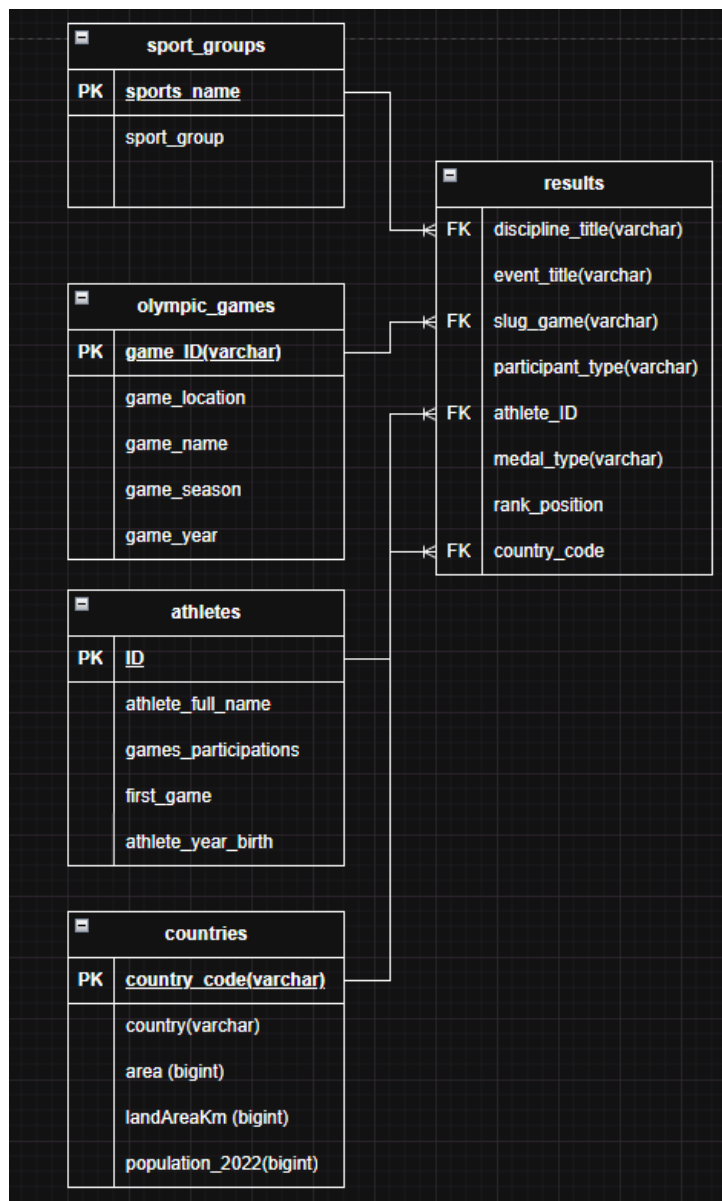


Figure 1: Relational Model of the Olympic Summer Games Database

2) Query: "Who got the most medals in these games?"

```
-- Find the Athlete with the highest total medal count
select c.country, r.medal_type, a.athlete_full_name, count(*) from results r
join countries c on r.country_code = c.country_code
join athletes a on a.id = r.athlete_ID
where r.medal_type is not null
group by c.country, r.medal_type, a.athlete_full_name
order by count(*) desc
limit 1;
```

3) Two questions that could be answered with this data

1. "Who are the top three athletes with the most **Gold** medals in the disciplines of Shooting, Triathlon, and Table Tennis?"
2. "Identify all athletes who won a medal in any Olympic Games edition held prior to the year 2008."

4) Two questions to check data quality

1. **Birth Year Validity:** "Are there any athletes with birth years that are clearly erroneous (e.g., in the future, or implying the athlete is under 10 or over 100 years old during the games)?"
2. **Medal Consistency:** "Do individual events award more than the expected number of medals (e.g., are there events with 5 Gold medals awarded but it is a single-player sport)?"

Database 2: Nutrition Information

1) Table Content and Data Quality Checks

In addition to checking for NULL descriptions in the `food_des` table, the following checks were performed to validate value ranges:

```
-- Check 1: Missing Descriptions (User Provided)
SELECT COUNT(*) as missing_descriptions
FROM food_des
WHERE long_desc IS NULL;

-- Check 2: Invalid Negative Nutrient Values
-- Nutrients cannot be negative.
SELECT COUNT(*) as negative_values
FROM nut_data
WHERE nutr_val < 0;

-- Check 3: Invalid Mass Values
-- Since values are per 100g, no component (protein, fat, etc.)
-- can exceed 100g.
SELECT ndb_no, nutr_no, nutr_val
FROM nut_data
JOIN nutr_def ON nut_data.nutr_no = nutr_def.nutr_no
WHERE units = 'g' AND nutr_val > 100;
```

2) ERM (Nutrition Database)

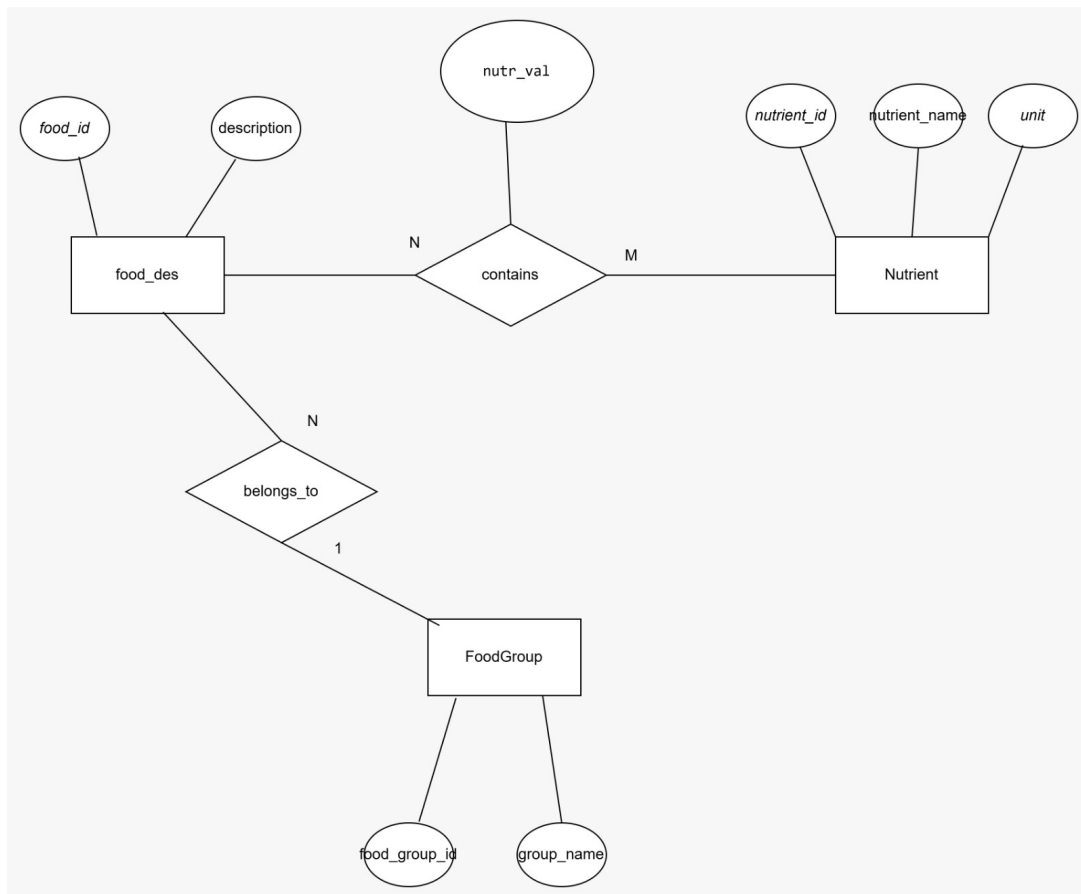


Figure 2: Entity Relationship Model of the Nutrition Database

3) Nutrients of Last Meal

```
SELECT f.long_desc AS food_name, n.nutrdesc AS nutrient,
       d.nutr_val AS amount, n.units
FROM food_des f
JOIN nut_data d ON f.ndb_no = d.ndb_no
JOIN nutr_def n ON d.nutr_no = n.nutr_no
WHERE
    f.long_desc ILIKE '%Apple%'
    OR f.long_desc ILIKE '%Cheddar Cheese%'
ORDER BY f.long_desc, n.nutrdesc;
```

4) Most Energy Rich Food Group

```
SELECT
    fg.fddrp_desc AS food_group,
    AVG(d.nutr_val) AS average_calories
FROM fd_group fg
JOIN food_des f ON fg.fdgrp_cd = f.fdgrp_cd
JOIN nut_data d ON f.ndb_no = d.ndb_no
JOIN nutr_def n ON d.nutr_no = n.nutr_no
WHERE n.nutrdesc ILIKE 'Energy%' AND n.units = 'kcal'
GROUP BY fg.fddrp_desc
ORDER BY average_calories DESC
```

```
LIMIT 1;
```

5) Good Sources of Vitamin C

```
SELECT
    f.long_desc AS food_name,
    d.nutr_val AS vitamin_c_amount,
    n.units
FROM food_des f
JOIN nut_data d ON f.ndb_no = d.ndb_no
JOIN nutr_def n ON d.nutr_no = n.nutr_no
WHERE
    n.nutrdesc ILIKE '%Vitamin_C%'
ORDER BY d.nutr_val DESC
LIMIT 20;
```